

CURRICULUM VITAE

RUNPENG LUO

CONTACT INFORMATION	Computer Science Building Department of Computer Science Princeton University Princeton, NJ, USA	Phone: (+01) 609 933 5742 E-mail: runpeng.luo@princeton.edu Website: runpengluo.github.io Google Scholar: Link
EDUCATION	Princeton University, Princeton, NJ, USA Doctor of Philosophy (Ph.D.) in Computer Science Sep 2024 - ongoing Master of Art (earned en passant) Sep 2024 - Dec 2026 Australian National University, Canberra, ACT, Australia Bachelor of Advanced Computing (Research and Development) (Honours) Feb 2020 - Dec 2023 First Class Honours - GPA: 6.813/7 Advisors: Dr. Yu Lin and Dr. Benjamin Schwessinger Strathfield South High School, Sydney, NSW, Australia High School Certificate Oct 2017 - Oct 2019 ATAR: 94.75	
EMPLOYMENT	Diversity Arrays Technology, Canberra, ACT, Australia R&D Solution Developer Feb 2024 - Aug 2024 Australian National University, Canberra, ACT, Australia Technical Assistant at the School of Biology Oct 2022 - Aug 2024 Summer Research Intern at the School of Computing 2021&2022 Summer Teaching Assistant COMP3320 High Performance Scientific Computation 2023 Semester 2 COMP7240 Introduction to Database Concepts 2023 Semester 1 COMP4300/8300 Parallel System 2023 Semester 1 BIOL8002 Advanced Topics in Quantitative Biology and Bioinformatics 2022 Semester 2 COMP2400/6240 Relational Database 2021&2022 Semester 2	
RESEARCH INTEREST	<i>Computational Biology</i> <i>Combinatorial Optimization</i> <i>High Performance Computation</i>	
RESEARCH EXPERIENCE	Graph Model and Algorithms for Haplotype-resolved Assembly on Dikaryotic Genome Feb 2023 - Dec 2023 Advisors: Dr. Benjamin Schwessinger, Dr. Lianrong Pu, and Dr. Qing Wang Objectives: Design and implement a bi-partition algorithm to assemble and phase dikaryotic genome using third-generation sequencing (TGS) data and Hi-C data. Plasmid Library Diversity Quantification using Nanopore Sequencing Data Jun 2023 - May 2024 Advisors: Dr. Joseph Brock and Dr. Benjamin Schwessinger Objectives: Design and implement a novel classification algorithm to quantify the plasmid combinations from the library mixture. De Novo Reconstruction of Viral Strains via Iterative Path Extraction from Assembly Graphs Dec 2021 - Oct 2022 Advisors: Dr. Yu Lin Objectives: Design and implement an assembly algorithm to reconstruct strains from viral quasispecies under De novo approach.	
AWARDS	ANU Summer Research Scholarship, 2021&2022 Summer, Canberra, Australia NSW Government School International Student Awards - Academic Achievement 2020, Sydney, Australia	

TALKS	NCI Informatics Technology for Cancer Research Meeting, (ITCR 2026), lightning talk, Cold Spring Harbor, NY, USA, Jul 2026. 27th Annual International Conference on Research in Computational Molecular Biology, (RECOMB 2023), Link, Istanbul, Turkey, Apr 2023.
RELEVANT SKILLS	Programing: Python, C/C++, Java, Haskell, Rust, ARMv7 Assembly, PostgreSQL Utilities: Bash, L^AT_EX, Anaconda, Git, VsCode, Jupyter Notebook
LANGUAGES	Mandarin Chinese (native) English (fluent)
REFERENCES	Dr. Benjamin Schwessinger <i>E-mail:</i> benjamin.schwessinger@anu.edu.au Associate Professor at the Research School of Biology, Australian National University Dr. Yu Lin <i>E-mail:</i> yu.lin@anu.edu.au Senior Lecturer at the School of Computing, Australian National University
PUBLICATIONS	Tam, R., Mareike, M., Luo, R., Luo, Z., Ashley, J., Sambasivam, P., John P, R., and Schwessinger, B., Long-read genomics reveal extensive nuclear-specific evolution and allele-specific expression in a dikaryotic fungus. <i>Genome Research</i>, (2025). 10.1101/gr.280359.124 Cleaver, A., Luo, R., Smith, O. B., Murphy, L., Schwessinger, B., and Brock, J., High-throughput optimisation of protein secretion in yeast via an engineered biosensor. <i>Trends in Biotechnology</i>, (2024). 10.1016/j.tibtech.2024.11.010 Luo, R. and Lin, Y., VStrains: De Novo Reconstruction of Viral Strains via Iterative Path Extraction From Assembly Graphs. <i>Proceedings of the 27th International Conference in Computational Molecular Biology (RECOMB 2023)</i>, 3-20 (2023). 10.1007/978-3-031-29119-7_1